

Case

1 what are the case functions syntactically?

2 How do we derive them? All of them?

3 From where do we derive them?

4 What is the semantics of case?

(1-4) from Bozsahin (2075)

① Case functions manifest categorial understanding of native speaker's understanding of who does what to whom.

- It is demonstration of understanding of competence for syntax and semantics of argument taking.
- All constructions can then assume all arguments are cased.

Example: an English ditransitive (3 args)

John gave Mary the book.

gave :: $(s/nP)/NP/NP : \lambda x \lambda y \lambda z. \text{give}'xyz$

— why $\text{give}'xyz$ rather than $\text{give}'xyz$?

John showed Mary herself/himself



gave :: (S \ NP) / NP / NP : $\lambda x \lambda y \lambda z. \text{give}' yxz$



- Three arguments. All of them must have case functions to demonstrate understanding of case by categories. (Actually, our understanding of speaker's ^{understanding} understanding)
 - Everything has case, as long as they are arguments.
 - wanted :: (S \ NP) / VP / NP believed :: (S \ NP) / S
- 

④ Semantics of case:

- For an argument a , its semantics is $\lambda p.p a$
- This means that case has an invariant (call it universal if you like—I don't.)
- However, it says more. It all depends on what kind of linguistic event P is.
- Proposal in the book: 3-way distinction is sufficient: events of ROOT, STEM, FINITE verbs.

For the running example, ALL 3 FORMS of the verb give assign DIFFERENT kind of cases, because the events are different:

1) I gave Mary the book
finite

2) I want to give Mary the book.
infinitive

3) I should give Mary the book
root

ACT

Refer to actual or realizable worlds.

Refer to possible worlds

Construct possible worlds.

WE HAVE TO CONSTRUCT TO BE ABLE TO REFER. ROOT FORM IS PRIMARY.

- The book claims that 3-way distinction in the semantics of P in $Op.p.a$ is SUFFICIENT for Case Typology.

- For the running example, we explain ALL of the following by syntacticizing the 3-way distinction of case.

- 1) I give / *give / * to give Mary the book.
- 2) I want / to give / *gives / *give / Mary the book.
- 3) I should give / *gives / *to give Mary the book.

(2-3) How do we derive case functions, and from where?

- Things to keep in mind:

- Although case semantics is invariant, obtaining $\lambda p.pa$ from a , what kind of linguistic event p corresponds to makes a difference for case.
- The syntactic counterpart of p must not be analytic; it must be ALREADY put together.

SYNTHE TIL

Categorization of case: (from page 95 of the book)

IT IS A RELATIONAL CATEGORY

1) Forward case functions

↓ Basic subcategorization

$$(\alpha, A:a) \rightarrow (\alpha, T/(T \setminus A) : \lambda p. pa)$$

- Kwakiwala show why α need not preserve lexical integrity (p. 125)

2) Back word case functions

$$(\alpha, A:a) \rightarrow (\alpha, T \setminus \underline{(T/A)} : \lambda p. pa)$$

↑ Basic subcategorization

Casing the arguments of a finite verb:

I gave Mary the book

- The source of case is the finite verb.
- More specifically, it is the category of the finite verb:

Basic subcategorization gave :: (S\NP)/NP/NP : $\lambda x \lambda y \lambda z$. give $y x z$
guarantees invariant semantics $\Leftarrow$  only this argument can
be cased from this category
(OUTERMOST)

gave :: $(S \backslash NP) / NP / NP : \lambda x \lambda y \lambda z. \text{give } y \text{ to } z$
 ↑ case

$$\frac{(S \backslash NP) / NP}{T} \backslash \frac{((S \backslash NP) / NP / NP)}{(T / A)} : \lambda p.p \circ$$

- Other subcategorizations must get their case functions from verbs where THEY are the OUTERMOST argument. THAT ENSURES INVARIANT semantics.

arrived :: $S \backslash NP : \lambda x. \text{arrive}' x$ → CASE $S / (S \backslash NP) : \lambda p.p \circ$
 $\overline{T} / \overline{T} \backslash \overline{A}$

- if we want invariant semantics to be sound, that is, only assign case for the correct argument,
- then it must have access to basic subcategorization to assign the right sort of case (finite form, root form, infinitive form)

arrive :: $s \backslash np : \lambda x. arrive' x \rightarrow s / (s \backslash np)$
: $\lambda p.p$ a

to arrive :: $vp : \lambda x. arrive' x \rightarrow ?$ no OUTERMOST

arrive :: $iv : \lambda x. arrive' x \rightarrow ?$ ARGUMENT

But then there are the transitives:

hits :: $(s \setminus np) / np : \lambda x \lambda y. hit' x y$ hit'

hit :: $n / np : \lambda x \lambda y. hit'' x y$ hit''

to hit :: $vp / np : \lambda x \lambda y. hit''' x y$ hit'''

Why same-looking semantics?
Same prominence relation, different event.

could be
different
depending on event type

- Casing a VP :

wants :: $(s \setminus \text{vp}) / \text{vp} : \lambda p \lambda x. \text{want}' (p x) x$



$\frac{(s \setminus \text{vp})}{\overline{T}} \setminus ((s \setminus \text{vp}) / \text{vp}) : \lambda p. \lambda o$
 $\setminus (T / A)$

- Casing an S :

believes :: $(s \setminus \text{vp}) / s : \lambda p \lambda x. \text{believe } p x$



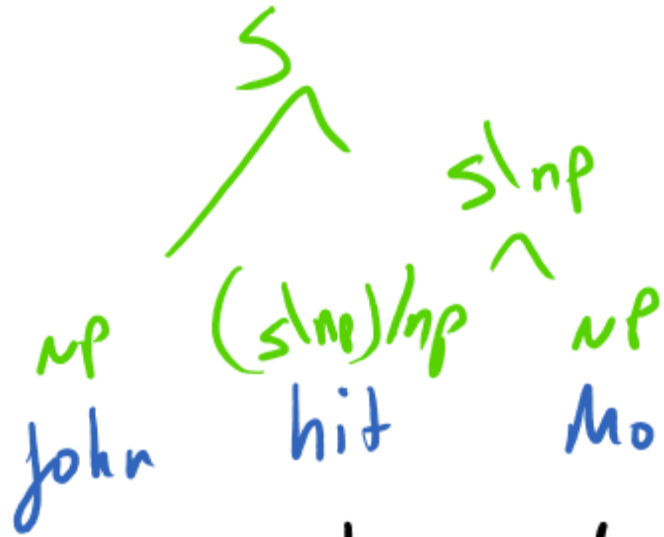
$\frac{(s \setminus \text{vp})}{\overline{T}} \setminus ((s \setminus \text{vp}) / s) : \lambda p. \lambda o$
 $\setminus (T / A)$

Case

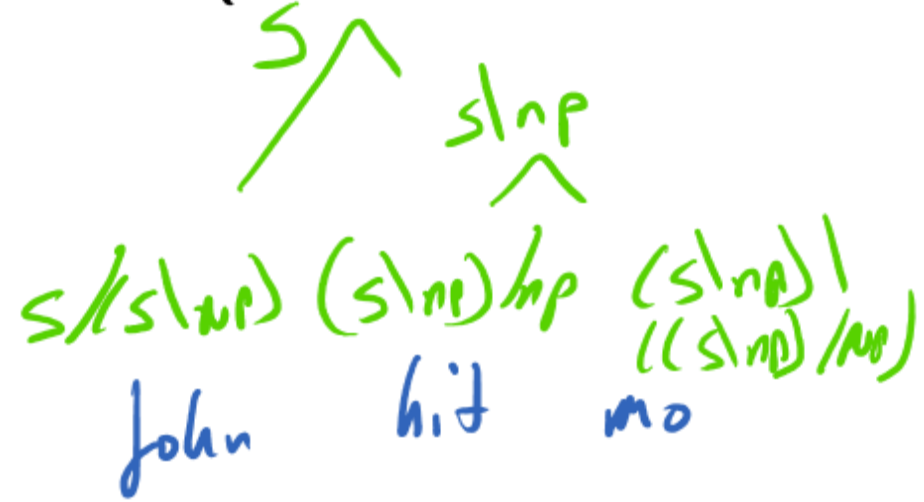
- ✓ 1 what are the case functions syntactically?
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- 5 WHAT IS ALL THAT FOR?

- Case functions can replace the uncased types if we have complete knowledge of case assigners.
- That gives an understanding of case.
(Knowing case presupposes knowing argument-taking)
- Moreover, we begin to PREDICT unseen structures from case functions.
- They give us discovery procedures.

1 not so interesting examples of analyses:



not cased



cased

2 More interesting syntactic structures expressing the same case relations:

John hit and Mary petted Mo

- John gave Mary and showed Harry the book.

- John gave Mary a book and Harry a flower.

- And we can now assume that all constructions assume their category expects cased arguments.

- why? Because constructions are case users; they are not case assigners. They are case to set THEIR semantics.

John gave Mary and showed Harry ...

$(s \backslash np) / np$



$(s \backslash np) / np / np ((s \backslash np) / np) ((s \backslash np) / np / np)$
gave Mary

John gave Mary a book and Harry a flower

$(S \setminus NP) / NP$



looks for a verbal category

even when
composed.

Why? because

all case

functions are

derived from

verbal

subcategorization.

(sub. Mary and a book)

$(S \setminus NP) \setminus ((S \setminus NP) / NP / NP)$

compose

$((S \setminus NP) / NP) \setminus ((S \setminus NP) / NP / NP)$

Mary

a book

Case

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- ✓ 5 WHAT IS ALL THAT FOR? DISCOVERY

- For use of case in the constructions, have a look at the lecture notes on, eh, constructions.

- The book claims that categorial crosslinguistic diversity requires the following distinction in category.

Dyirbal

IV_{abs}
VP_{abs}

Inuit

IV_{erg}
VP_{erg}

Icelandic

IV_{acc}
VP_{acc}

Latin

IV_{nom}
VP_{nom}

Tagalog/English

IV
VP

Back to explaining the following by case ALONE:

- Mary of ① : $(\langle \langle \text{S} \backslash \text{np} \rangle / \text{np} \rangle \backslash (\langle \langle \text{S} \backslash \text{np} \rangle / \text{np} / \text{np} \rangle) : \text{p.p. Mary}'$
 - Mary of ② : $(\text{np} / \text{np}) \backslash ((\text{vp} / \text{np}) / \text{np}) : \text{p.p. Mary}'$
 - Mary of ③ : $(\text{IV} / \text{np}) \backslash ((\text{IV} / \text{np}) / \text{np}) : \text{p.p. Mary}'$
- (p's are different depending of structure)

- ↓ root
- 1) I give / *give / *to give Mary the book.
 - 2) I want /to give / *gives / *give / Mary the book.
 - 3) I should give / *gives / *to give Mary the book.